

Skills Summary

Writing Inequalities from Verbal Constraints

Use this reference while completing your worksheet or when studying.

1. Turn a phrase into an inequality symbol

The boundary question: is the number in the sentence itself allowed?

- *more than, over, greater than* $\rightarrow >$ (boundary not allowed — open circle)
- *less than, under, fewer than* $\rightarrow <$ (open circle)
- *at least, no less than, minimum of* $\rightarrow \geq$ (boundary allowed — closed circle)
- *at most, no more than, maximum of* $\rightarrow \leq$ (closed circle)

Say what the variable counts *in words* first — that sentence is part of the answer.

Example: A theatre seats no more than 250 people. Write an inequality for the size s of an audience.

Step 1. “No more than” asks whether 250 itself is allowed — it is, since a full house of 250 is legal, so this is the closed direction.

Step 2. The audience size is the quantity being limited, and the limit is above it, so s sits on the small side of 250.

$$s \leq 250 \quad (\text{closed circle at 250, shaded left})$$

Try it: You must be at least 16 years old to get a learner’s permit. Write an inequality for the ages a that qualify.

$$\text{check: } 16 \leq a$$

2. Build the inequality from a story

Most story constraints have the same three pieces:

(one-time amount) + (amount each time) $\times$ (how many) compared with (the limit)

The number attached to the variable is the one that happens *again and again*; the number standing alone happens *once*. Getting those two the right way round is most of the work.

Example: A \$48 gift card, a \$12 snack, and rentals at \$9 each. How many rentals r ?

Step 1. The snack is bought once and the rentals repeat, so 9 is the number that multiplies r and 12 stands alone.

Step 2. Spending cannot pass the card’s value: $12 + 9r \leq 48$, so $9r \leq 36$ and dividing by 9 gives the count.

$$r \leq 4 \quad (\text{check: } 12 + 9 \cdot 4 = 48 \text{ exactly})$$

Try it: A fair charges \$8 to get in and \$3 per ride. You have at most \$35. Write and solve an inequality for the number of rides n .

$$\text{check: } 8 + 3n \leq 35$$

3. Solve, flip when you must, and read it back

Solving an inequality works exactly like solving an equation, with *one* extra rule:

Multiplying or dividing both sides by a negative number reverses the symbol.
Adding and subtracting never do.

Then **check one value** from your answer in the original sentence, and finally say what the answer means for the situation — “at most 4 rentals” rather than just “ $r \leq 4$ ”.

Example: A tank holds 300 litres and drains 25 litres a minute. When is more than 100 litres left?

Step 1. The amount left shrinks, so it is $300 - 25t$; “more than” is strict, which makes this a $>$ problem with a negative coefficient waiting.

Step 2. $300 - 25t > 100$ gives $-25t > -200$, and dividing by -25 flips the symbol.

$t < 8$ (check $t = 4$: 200 litres left, which is more than 100)

Try it: A balloon is at 180 metres and descends 20 metres a minute. For which times t is it above 60 metres?

$9 > t$:xəpə

(!) Watch out: the flip rule is about *multiplying or dividing* by a negative, never about subtracting one. And a solved inequality is not yet an answer to a story: if the variable counts objects, finish by saying how many whole objects are possible.